

# TRAPNET Core Concepts & Learning Mastery

*Complete guide covering Microcontrollers, Edge AI, Multi-Modal Sensor Fusion, Power Electronics, and Wise AI Assistant Collaboration Tactics.*

## 1. Embedded Microcontrollers & Real-Time Firmware (ESP32)

- **FreeRTOS Dual-Core Multitasking:** ESP32 WROOM-32 houses dual 32-bit Tensilica LX6 cores @ 240MHz. Core 0 manages mmWave radar & GSM UART queues; Core 1 handles real-time interrupts & relay triggers.
- **GPIO Dynamics & Boot Pin Restrictions:** GPIO34, 35, 36, 39 are input-only pins requiring 10k external pull-down resistors. Boot pins (GPIO0, 2, 12, 15) control boot modes; GPIO12 must stay LOW at boot.
- **Hardware Serial/UART Buses:** UART2 (GPIO16/17 @ 256000 baud) for mmWave radar binary frames; SoftSerial UART1 (GPIO26/27 @ 9600 baud) for SIM800L GSM AT commands.
- **Hardware Interrupts & Debouncing:** IR beam break & SW-420 vibration sensors trigger non-blocking hardware edge interrupts instead of CPU polling loops.

## 2. Radar Physics & Multi-Modal Sensor Fusion

- **24GHz FMCW mmWave Radar Principles:** HLK-LD2410C transmits FMCW signals at 24GHz. Detects static human breathing & micro-motion energy gates up to 6 meters.
- **Spatial Vision-Radar Fusion Formula:** Cross-verifies camera optical distance  $D_{vision} = (f * H) / h$  against mmWave radar distance  $D_{radar}$ . If  $|D_{vision} - D_{radar}| \leq 25cm$ , threat is confirmed.
- **False Alarm Rejection:** Multi-modal energy gating rejects environmental noise from wind, foliage, pets, and rain.

## 3. Edge AI & Mobile Computer Vision (YOLOv8)

- **YOLOv8 Architecture:** Anchor-free split head detection model predicting bounding boxes and class labels in a single pass.
- **INT8 Model Quantization:** Post-Training Quantization converts FP32 weights to INT8, shrinking model size by 75% and accelerating NPU inference by 4x-5x.
- **Smartphone NPU Acceleration:** TFLite with NNAPI delegate executes model on mobile Hexagon NPU / GPU, reducing latency from 150ms down to sub-15ms.

## 4. How to Use AI (Antigravity/Gemini) Wisely as an Engineer

- **Provide Complete Hardware Context:** Specify exact pin maps, baud rates, voltage levels, and library dependencies in your prompts.
- **Use Planning Mode First:** Request system block diagrams, protocol tables, and error handling strategies before writing code.
- **Paste Raw Stack Traces:** Provide un-truncated compiler errors so the AI can diagnose line-by-line root causes.
- **Step-by-Step Empirical Verification:** Compile and test modules on real hardware before building full system integration.